Pre-lab

Name: Colby Alley

Lab Partner: Alek Presley

PL1: From the diode data sheet determine the following values: peak repetitive reverse voltage, average rectified forward current, non-repetitive peak forward current. After completing the prelab calculations and simulations (i.e. PL2 to PL5) check the results against these values to verify that the diode specifications are not exceeded.

• From the diode's (1N4001) data sheet:

Peak Repetitive Reverse Voltage $V_{RRM} = \boxed{50.0 \text{ V}}$

Average Rectified Forward Current $I_O = \boxed{1.0 \text{ A}}$

(single phase, resistive load, 60 Hz, $T_A = 75°\text{C}$)

Non-Repetitive Peak Surge Current $I_{FSM} = \boxed{30.0 \text{ A}}$ (for 1 cycle)

*Note after returning from doing PL2 to PL5: None of the values are exceeded from future computations.

PL2: Calculate V_m for a transformer which has an output RMS voltage equal to 14 V. Calculate V_{DC} and V_R for the cases (a) without and (b) with the filter capacitor (Fig. 1 and Fig. 3). The frequency is 60 Hz. Assume the diode drop to be $V_\gamma = 0.7$ V. Use the filter capacitor and load resistance values given at the beginning of the Lab Procedure section. For case (b) calculate the peak voltage across the capacitor.

• Let the transformer voltage be $v(t) = V_m \sin(\omega t)$:

where $\omega = 2\pi f$ and $f = 60$ Hz... Recall $V_{rms} = 14$ V and $V_m = \sqrt{2}V_{rms}$.

$$V_m = \sqrt{2} \cdot 14 \approx \boxed{19.799 \text{ V}} \approx 20.0 \text{ V}.$$

Case A) without capacitor: ($\theta = 2\pi f$)

$$V_{DC} = \frac{1}{2\pi}\int_0^{2\pi} V_m \sin(\theta) d\theta$$
$$V_{DC} = \frac{1}{2\pi}\left(\int_0^{\pi} V_m \sin(\theta) d\theta + \int_{\pi}^{2\pi} 0 d\theta\right)$$
$$V_{DC} = \frac{V_m}{2\pi}\Big[-\cos(\theta)\Big]_0^{\pi}$$
$$V_{DC} = \frac{V_m}{2\pi}\left[-\cos(\pi) - (-\cos(0))\right]$$
$$V_{DC} = \frac{V_m}{2\pi}[1+1]$$
$$V_{DC} = \frac{V_m}{\pi}$$
$$V_{DC} = \frac{V_m - V_\gamma}{\pi}$$

Recall $V_\gamma = 0.7$ V...

$$V_{DC} = \frac{19.799 - 0.7}{\pi} \approx \boxed{6.079 \text{ V.}}$$

for the ripple voltage without capacitor is just $V_R = V_m - V_\gamma$.

$$V_R = V_m - V_\gamma$$

$$V_R = (19.799 - 0.7) \approx \boxed{19.099 \text{ V.}}$$

Case B) with capacitor:

$$V_R = \frac{(V_m - V_\gamma)}{f R_L C}$$

$$V_{DC} = (V_m - V_\gamma) - \frac{V_R}{2}$$

$$R_L = 1 \text{ k}\Omega, V_\gamma = 0.7 \text{ V}, V_m = 19.799 \text{ V}, f = 60 \text{ Hz}, C = 47 \ \mu\text{F}.$$

$$V_R = \frac{(19.799 - 0.7)}{(60)(1 \cdot 10^3)(47 \cdot 10^{-6})} \approx \boxed{6.773 \text{ V.}}$$

$$V_{DC} = (19.799 - 0.7) - \frac{6.773}{2} \approx \boxed{15.713 \text{ V.}}$$

Peak voltage across the capacitor:

$$V_C = V_m - V_\gamma$$

$$V_C = 19.799 - 0.7 \approx \boxed{19.099 \text{ V.}}$$

PL3: Estimate the average load current and the diode's peak current (Bottom pane of Figure 4).

Case A.) Without filter capacitor:

$$I_{Lav} = \frac{V_{DC}}{R_L}$$

$$R_L = 1 \text{ k}\Omega,\ V_{DC} = 6.079 \text{ V},\ V_m = 19.779 \text{ V},\ V_\gamma = 0.7 \text{ V}.$$

$$I_{Lav} = \frac{6.079}{1 \cdot 10^3} \approx 6.079 \cdot 10^{-3} \approx \boxed{6.079 \text{ mA.}}$$

$$I_{DP} = \frac{V_m - V_\gamma}{R_L}$$

$$I_{DP} = \frac{19.779 - 0.7}{1 \cdot 10^3} \approx 19.079 \cdot 10^{-3} \approx \boxed{19.079 \text{ mA.}}$$

Case B.) With filter capacitor:

$$V_{DC} = 15.713 \text{ V (from above).}$$

$$I_{Lav} = \frac{V_{DC}}{R_L}$$

$$I_{Lav} = \frac{15.713}{1 \cdot 10^3} \approx 15.713 \cdot 10^{-3} \approx \boxed{15.713 \text{ mA.}}$$

$$V_R = 6.773 \text{ V (from above).}$$

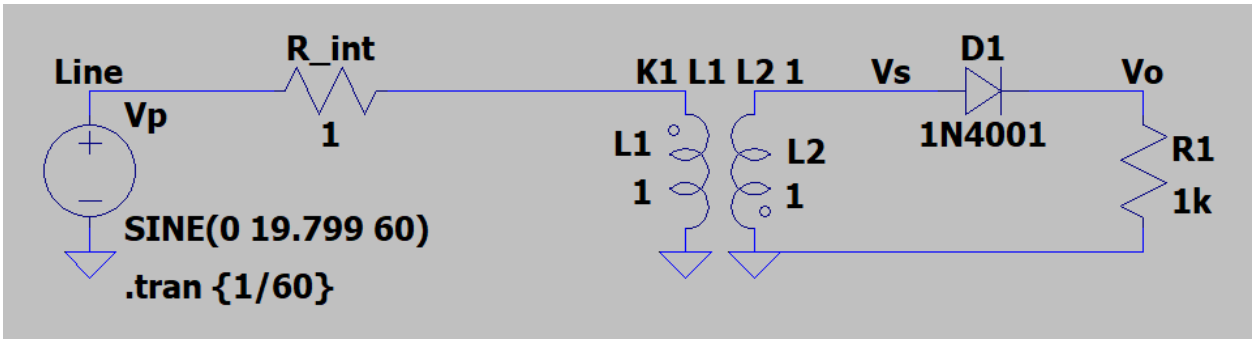
$$I_{DP} = \frac{(V_m - V_\gamma - V_R)}{R_L} + \frac{2\pi(V_m - V_\gamma)}{R_L}\sqrt{\frac{2V_m}{V_R}}$$

$$I_{DP} = \frac{(19.799 - 0.7 - 6.773)}{1 \cdot 10^3} + \frac{2\pi(19.799 - 0.7)}{1 \cdot 10^3}\sqrt{\frac{(2)(19.799)}{6.773}}$$

$$I_{DP} \approx 302.479 \cdot 10^{-3} \approx \boxed{302.479 \text{ mA.}}$$

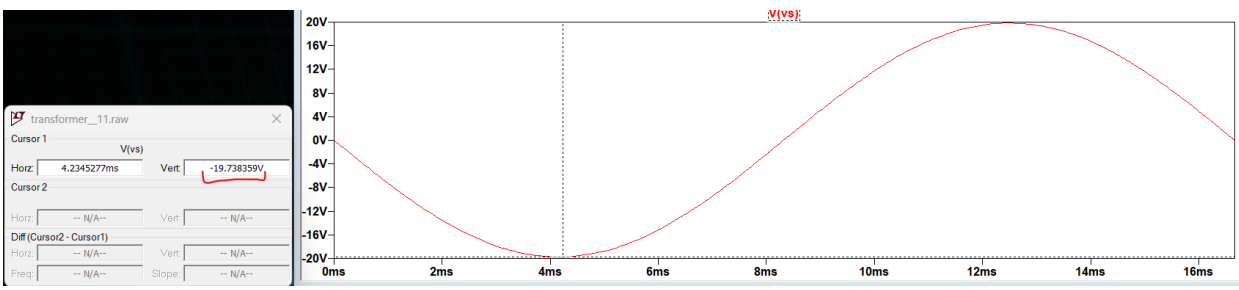
PL4: Verify your estimates in PLl and PL2 using SPICE simulations. In the SPICE simulation, model the transformer the way taught in the class.

Part 1: Without capacitor verification in PL1 and PL2



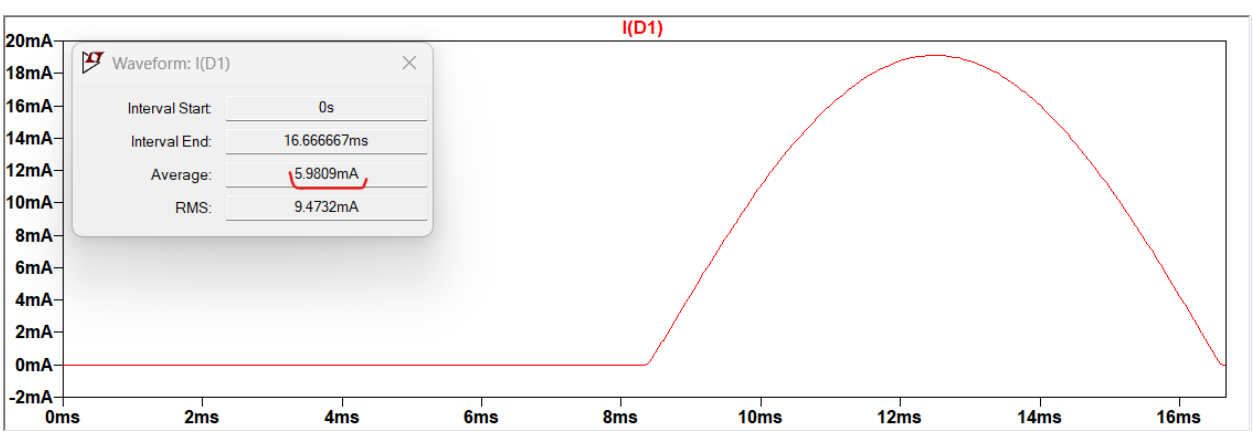
Circuit without capacitor and with transformer for one period

For PL1



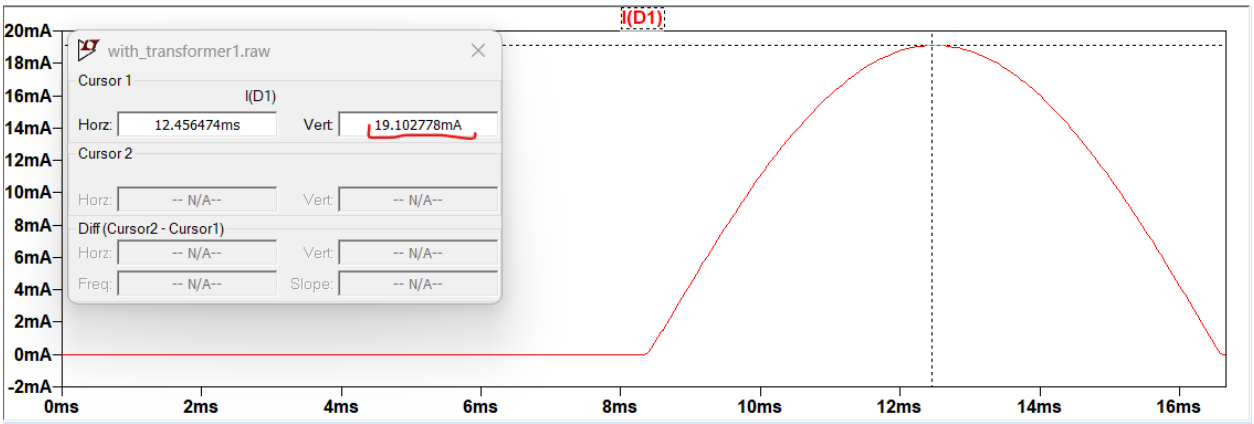
Circuit waveform for one period verifying V_{RMM} of diode (Peak Repetitive Reverse Voltage)

Since $V_{RMM} = 50$ V and the simulation states $\boxed{-19.738 \text{ V}}$, the diode will not exceed the limit rating.



Circuit waveform for one period verifying I_O of diode (Average Rectified Forward Current)

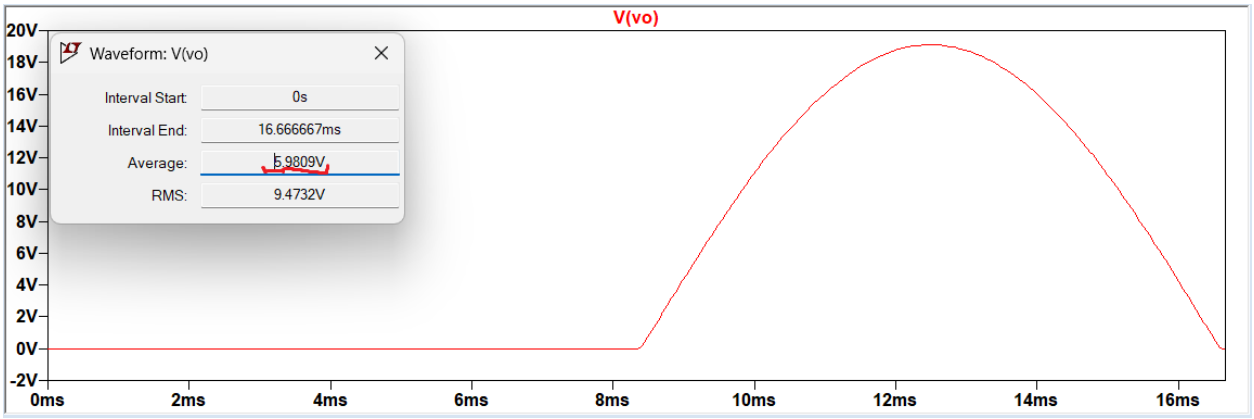
Since $I_O = 1.0$ A and the simulation states $\boxed{5.981 \text{ mA}}$, the diode will not exceed the limit rating.



Circuit waveform for one period verifying I_{FSM} of diode (Non-repetitive Peak Forward Current)

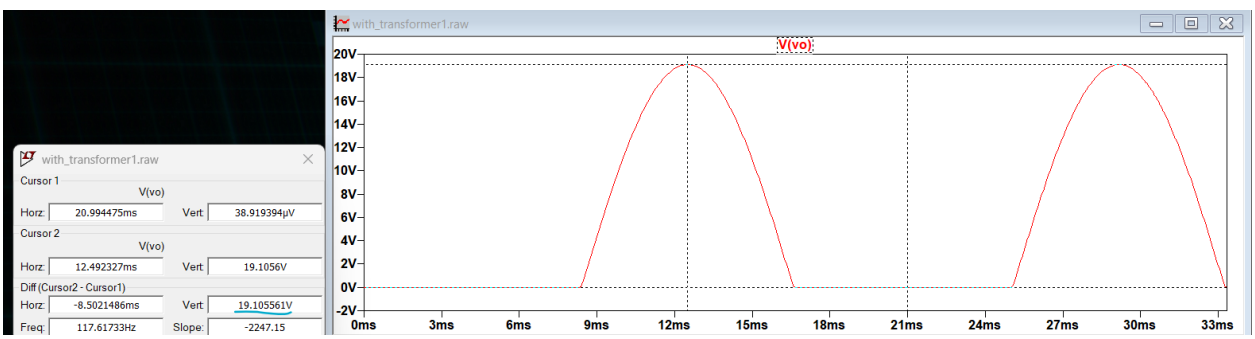
Since $I_{FSM} = 30.0$ A and the simulation states $\boxed{19.103 \text{ mA}}$, the diode will not exceed the limit rating. This is also similar to computation value.

For PL2



Circuit waveform for one period verifying V_{DC} (DC component)

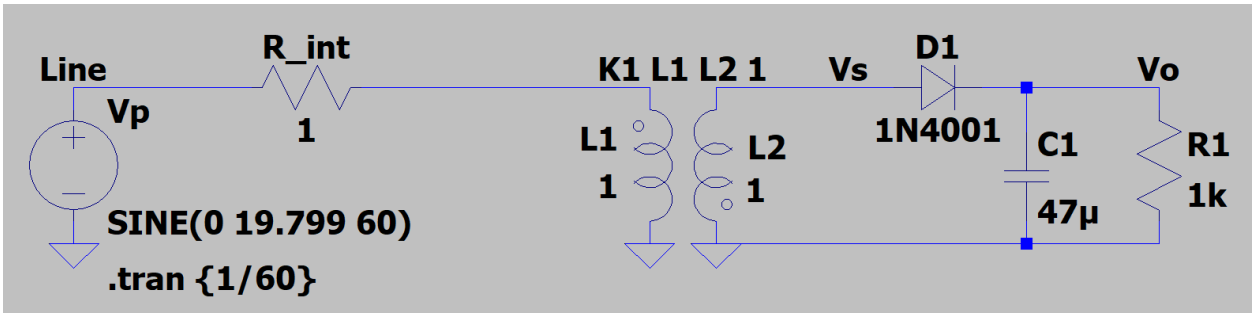
$$V_{DC} = \boxed{5.9809 \text{ V}} \approx \boxed{6.079 \text{ V.}}$$



Circuit waveform with two periods verifying V_R (ripple voltage)

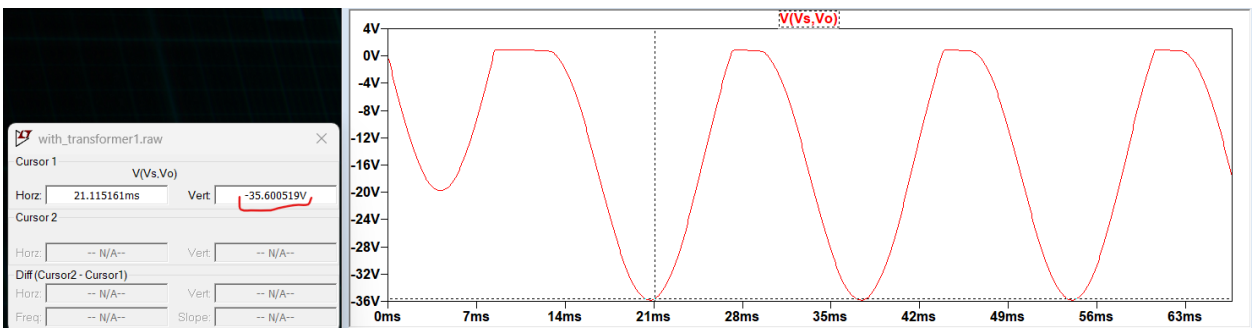
$$V_R = \boxed{19.099 \text{ V}} \approx \boxed{19.106 \text{ V.}}$$

Part 2: With capacitor verification in PL1 and PL2



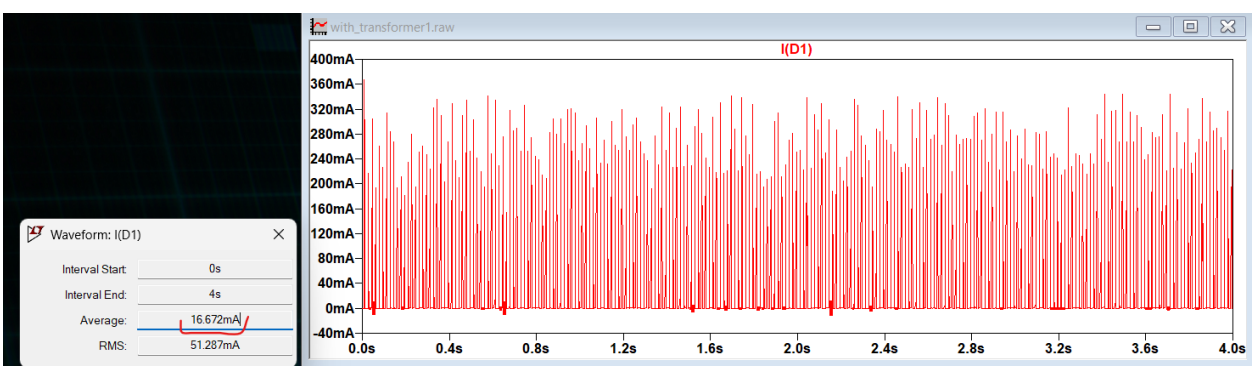
Circuit with capacitor and with transformer for one period

For PL1



Circuit waveform for four periods verifying V_{RMM} of diode (Peak Repetitive Reverse Voltage)

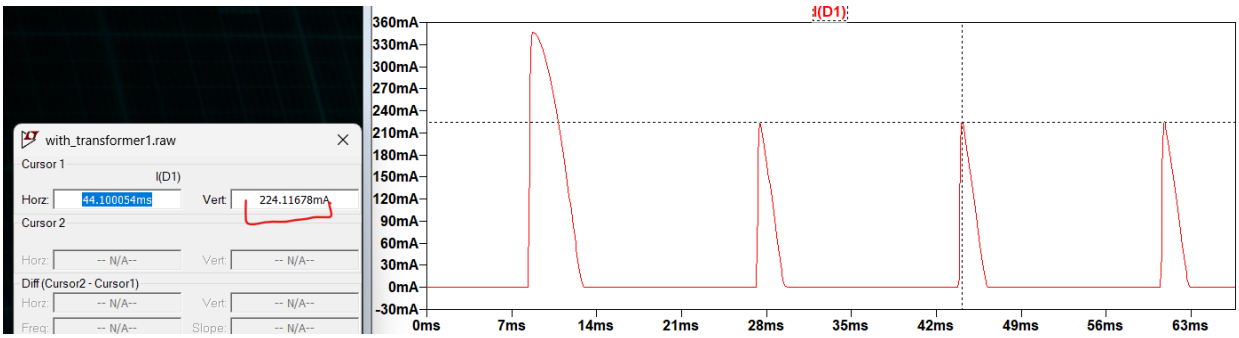
Since $V_{RMM} = 50$ V and the simulation states $\boxed{-35.601 \text{ V}}$ (after the capacitor has reached steady state), the diode will not exceed the limit rating.



Circuit waveform for 240 periods (4 seconds) verifying I_O of diode (Average Rectified Forward Current)

Since $I_O = 1.0$ A and the simulation states $\boxed{16.672 \text{ mA}}$ (after the capacitor has reached

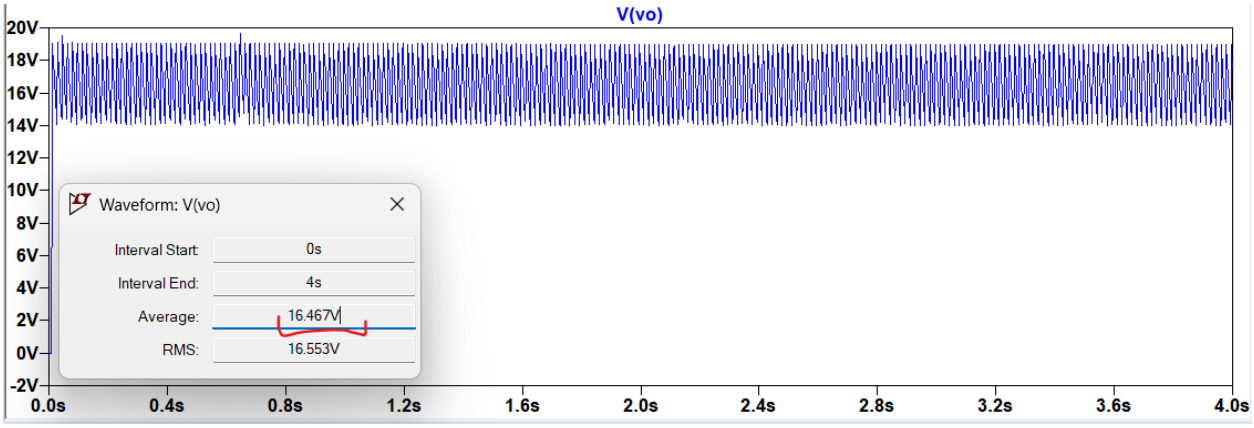
steady state), the diode will not exceed the limit rating. This is at a four second interval. The less cycles, the less accuracy determined.



Circuit waveform for four periods verifying I_{FSM} of diode (Non-repetitive Peak Forward Current)

Since $I_{FSM} = 30.0$ A and the simulation states $\boxed{224.117 \text{ mA}}$ (after the capacitor has reached steady state) the diode will not exceed the limit rating. Please note: as this approaches a higher time interval, it will become closer to the computed diode peak value of 302.479 mA.

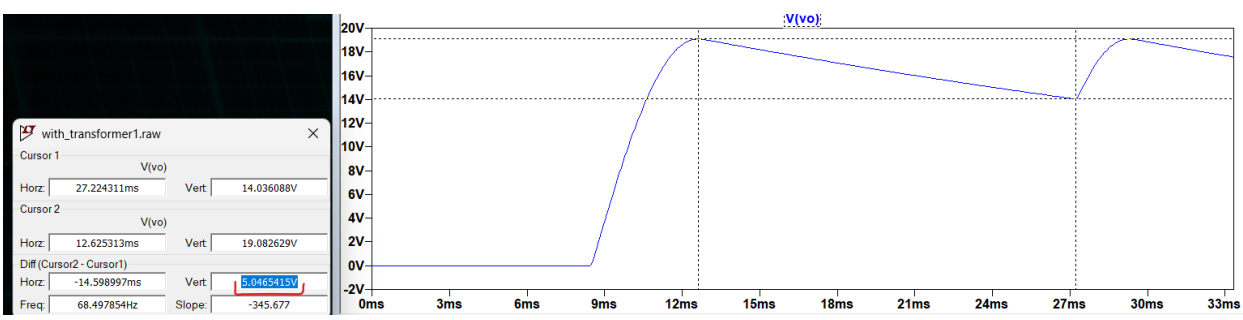
For PL2



Circuit waveform for 240 periods (4 seconds) verifying V_{DC} (DC component)

$$V_{DC} = \boxed{15.713 \text{ V}} \approx \boxed{16.467 \text{ V.}}$$

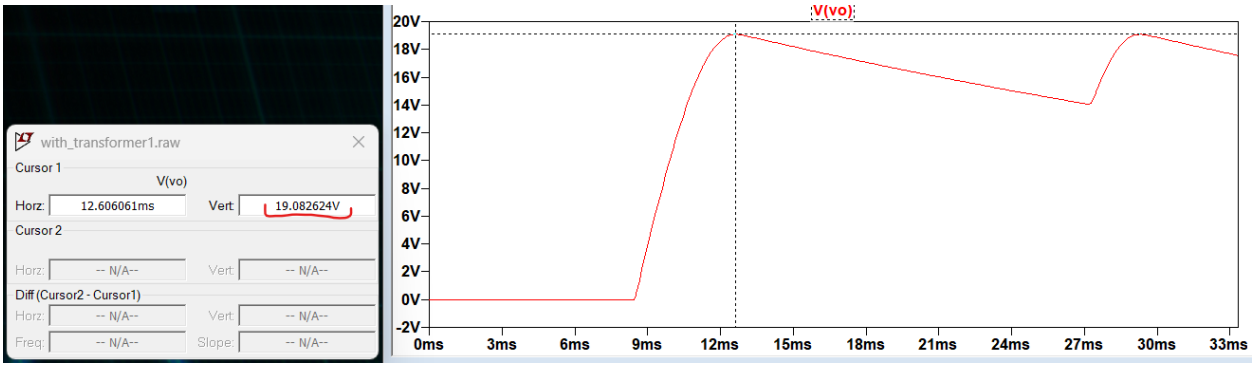
LTspice software is more accurate. The LTspice is spanned to a four second interval to allow for closer accurate (240 periods). The less periods, the less accurate the measured would be. Again, the lab equation is based off V_R, which is more of an approximation.



Circuit waveform for two periods verifying V_R (ripple voltage)

$$V_R = \boxed{6.773 \text{ V}} \approx \boxed{5.047 \text{ V.}}$$

Again, LTspice software is more accurate. The lab equation is more an approximation.



Circuit waveform for two periods verifying V_C (peak of capacitor)

$$V_C = \boxed{19.099 \text{ V}} \approx \boxed{19.083 \text{ V.}}$$

PL5: For the circuit with the filter capacitor, repeat PL3 for the cases (a) with a load resistance of 195 Ω and (b) with a load resistance of 4.7 kΩ.

• Case A) 195 Ω (two 390 Ω in parallel)

without filter capacitor (not required but for reference):

$$I_{L,av} = \frac{V_{DC}}{R_L}$$

$$R_L = 195\ \Omega,\ V_{DC} = 6.079 \text{ V},\ V_m = 19.779 \text{ V},\ V_\gamma = 0.7 \text{ V}.$$

$$I_{L,av} = \frac{6.079}{195} \approx 31.174 \cdot 10^{-3} \approx \boxed{31.174 \text{ mA.}}$$

$$I_{DP} = \frac{V_m - V_\gamma}{R_L}$$

$$I_{DP} = \frac{19.779 - 0.7}{195} \approx 97.841 \cdot 10^{-3} \approx \boxed{97.841 \text{ mA.}}$$

with filter capacitor (re-compute V_{DC} and V_R):

$$V_R = \frac{(V_m - V_\gamma)}{f R_L C}$$

$$V_R = \frac{19.779 - 0.7}{(60)(195)(47 \cdot 10^{-6})} \approx \boxed{34.695 \text{ V.}}$$

$$V_{DC} = (V_m - V_\gamma) - \frac{V_R}{2}$$

$$V_{DC} = (19.779 - 0.7) - \frac{34.695}{2} \approx \boxed{1.732 \text{ V.}}$$

$$I_{L,av} = \frac{V_{DC}}{R_L}$$

$$I_{L,av} = \frac{1.732}{195} \approx \boxed{8.882 \text{ mA.}}$$

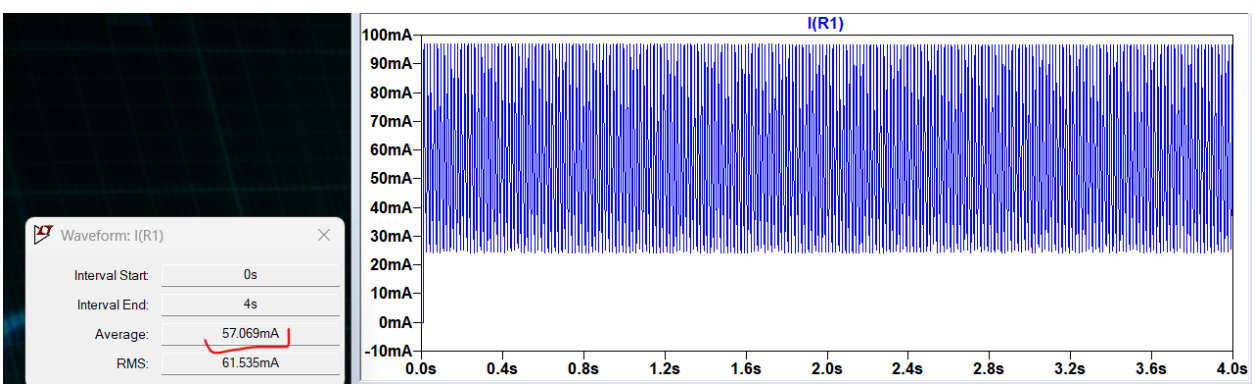
$$I_{DP} = \frac{(V_m - V_\gamma - V_R)}{R_L} + \frac{2\pi(V_m - V_\gamma)}{R_L}\sqrt{\frac{2V_m}{V_R}}$$

$$I_{DP} = \frac{(19.779 - 0.7 - 34.695)}{195} + \frac{2\pi(19.779 - 0.7)}{195}\sqrt{\frac{(2)(19.779)}{34.695}}$$

$$I_{DP} \approx \boxed{576.342 \text{ mA.}}$$

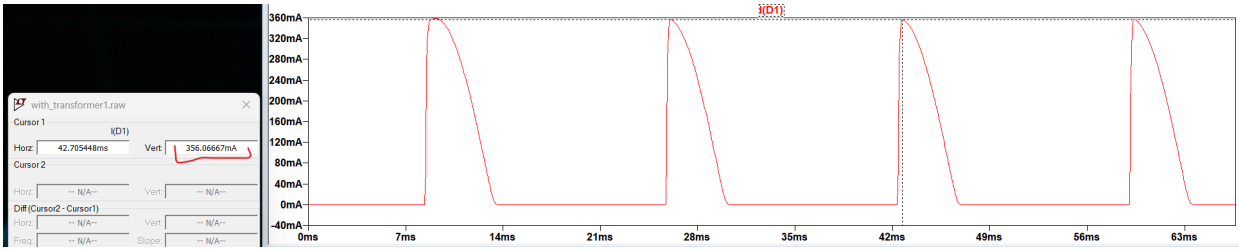
*So far, based off computation and using information from the data sheet, there is no limit rating exceeded.

LTspice simulation data (for circuit with capacitor only):



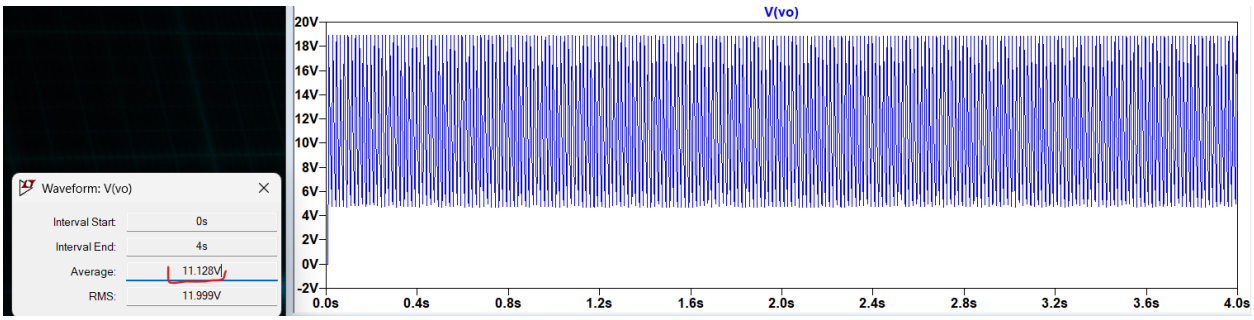
Circuit waveform for 240 periods (4 seconds) to measure $I_{L,av}$ (Average Load Current)

$I_{L,av}$ from LTspice $\approx$ $\boxed{57.069 \text{ mA.}}$



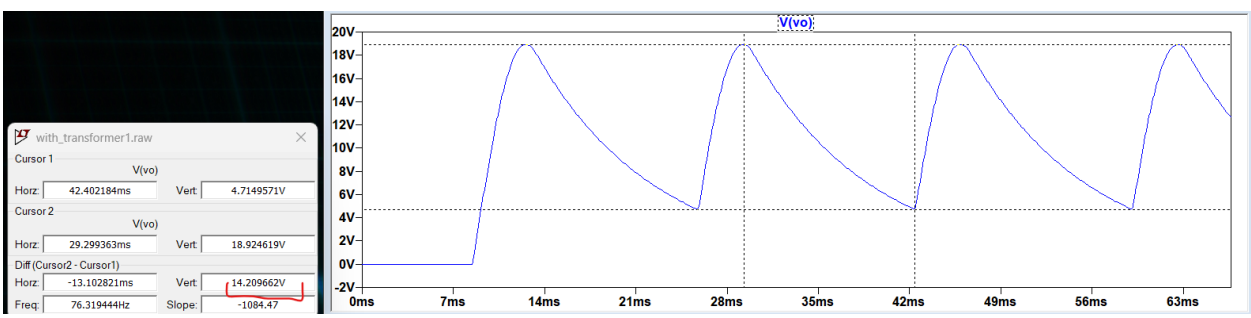
Circuit waveform for four periods to measure I_{DP} (Peak Diode Current)

I_{DP} from LTspice $\approx$ $\boxed{356.067 \text{ mA.}}$



Circuit waveform for 240 periods (4 seconds) to measure V_{DC} (DC component)

V_{DC} from LTspice $\approx$ $\boxed{11.128 \text{ V.}}$



Circuit waveform for four periods to measure V_R (Ripple voltage)

V_R from LTspice $\approx$ $\boxed{14.210 \text{ V.}}$

- Case B) 4.7 kΩ without filter capacitor (not required but for reference):

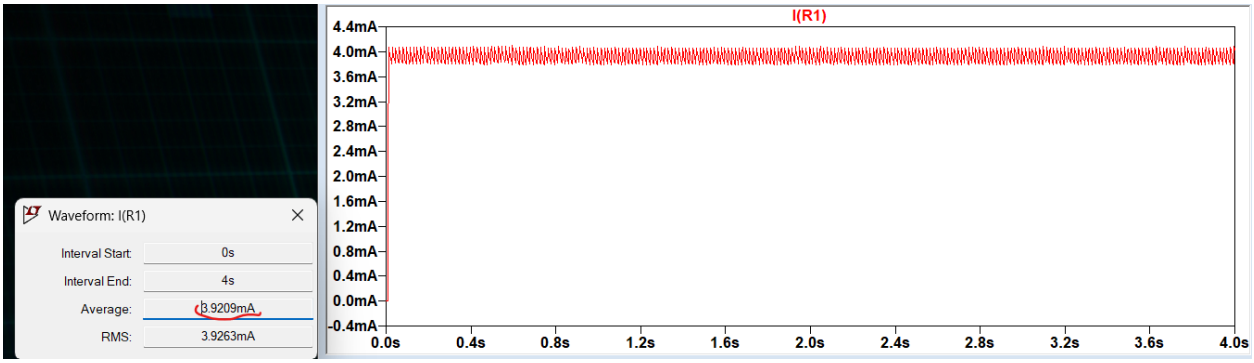
$$
\begin{aligned}
I_{L,av} &= \frac{V_{DC}}{R_L} \\
R_L &= 4.7\ \text{k}\Omega,\ V_{DC} = 6.079\ \text{V},\ V_m = 19.779\ \text{V},\ V_\gamma = 0.7\ \text{V}. \\
I_{L,av} &= \frac{6.079}{4.7 \cdot 10^3} \approx 1.293 \cdot 10^{-3} \approx \boxed{1.293\ \text{mA.}} \\
I_{DP} &= \frac{V_m - V_\gamma}{R_L} \\
I_{DP} &= \frac{19.779 - 0.7}{4.7 \cdot 10^3} \approx 4.059 \cdot 10^{-3} \approx \boxed{4.059\ \text{mA.}}
\end{aligned}
$$

with filter capacitor (re-compute V_{DC} and V_R):

$$
\begin{aligned}
V_R &= \frac{(V_m - V_\gamma)}{f R_L C} \\
V_R &= \frac{19.779 - 0.7}{(60)(4.7 \cdot 10^3)(47 \cdot 10^{-6})} \approx \boxed{1.439\ \text{V.}} \\
V_{DC} &= (V_m - V_\gamma) - \frac{V_R}{2} \\
V_{DC} &= (19.779 - 0.7) - \frac{1.439}{2} \approx \boxed{18.364\ \text{V.}} \\
I_{L,av} &= \frac{V_{DC}}{R_L} \\
I_{L,av} &= \frac{18.364}{4.7 \cdot 10^3} \approx \boxed{3.907\ \text{mA.}} \\
I_{DP} &= \frac{(V_m - V_\gamma - V_R)}{R_L} + \frac{2\pi(V_m - V_\gamma)}{R_L}\sqrt{\frac{2V_m}{V_R}} \\
I_{DP} &= \frac{(19.779 - 0.7 - 1.439)}{4.7 \cdot 10^3} + \frac{2\pi(19.779 - 0.7)}{4.7 \cdot 10^3}\sqrt{\frac{(2)(19.779)}{1.439}} \\
I_{DP} &\approx \boxed{137.482\ \text{mA.}}
\end{aligned}
$$

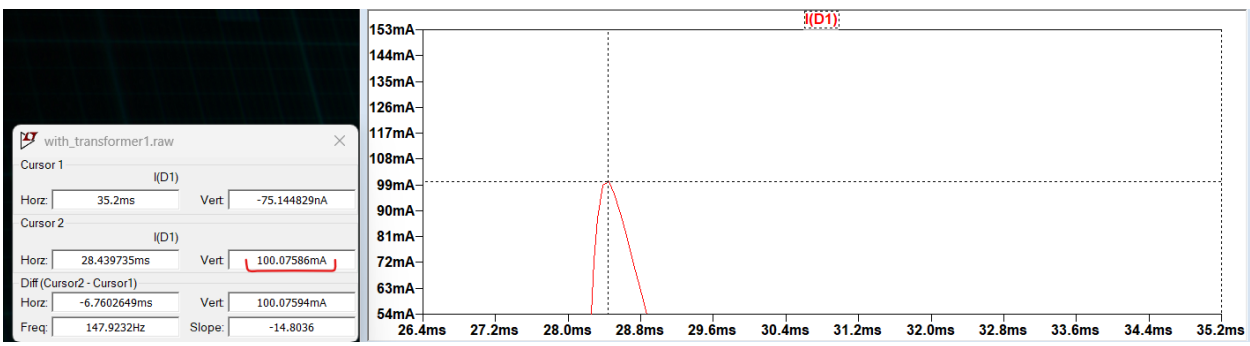
*Again, based off computation and using information from the data sheet, there is no limit rating exceeded.

LTspice simulation data (for circuit with capacitor only):



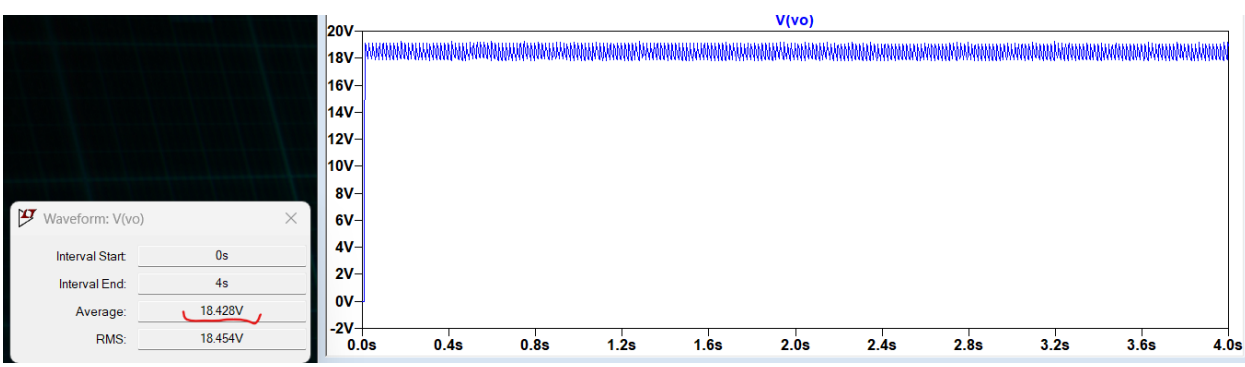
Circuit waveform for 240 periods (4 seconds) to measure $I_{L,av}$ (Average Load Current)

$I_{L,av}$ from LTspice $\approx$ $\boxed{3.921 \text{ mA.}}$



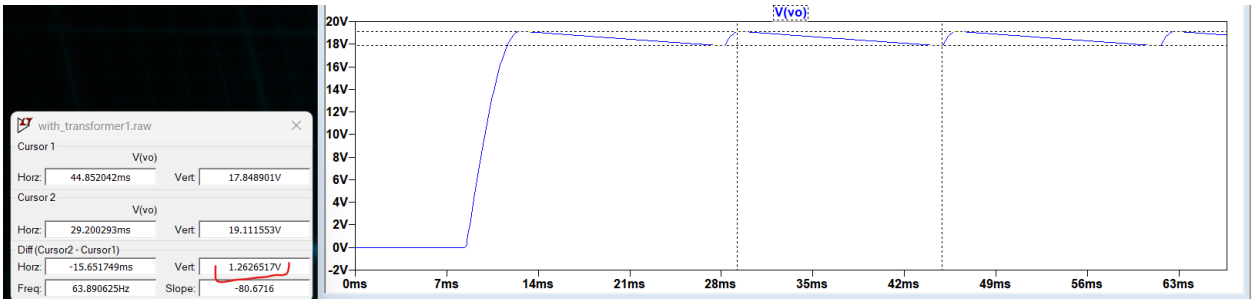
Circuit waveform for four periods to measure I_{DP} (Peak Diode Current)

I_{DP} from LTspice $\approx$ $\boxed{100.076 \text{ mA.}}$



Circuit waveform for 240 periods (4 seconds) to measure V_{DC} (DC component)

V_{DC} from LTspice $\approx$ $\boxed{18.428 \text{ V.}}$



Circuit waveform for four periods to measure V_R (Ripple voltage)

V_R from LTspice $\approx$ $\boxed{1.263 \text{ V.}}$